

Discretionary Decisions Catalogue

Dissertation: A Cloud-Based Analytics Engineering Platform for Public-Sector Accountability:
Deprivation and Local Authority Spending in England

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About this catalogue

Building a data platform involves many small choices that never show up in the final charts: which measure to use, which authorities to leave out, how to deal with gaps. This catalogue records the choices that mattered, as Objective 3 of the dissertation requires. For each one it sets out what was decided, what else could have been done, why this option was chosen and what difference it made. It also notes which part of Busuioc's (2021) accountability framework (Information, Explanation or Consequences) the decision touches.

Some decisions and interpretations changed as the work went on. In those cases the original reasoning has been kept alongside the correction and marked "Revised", so the record shows how the analysis developed and not just where it ended up.

The proposal expected seven categories of decision. In practice the analysis kept raising new questions, so the catalogue grew well beyond that. Sections 9.3 and 9.4 of the dissertation discuss this.

Entries are numbered in the order they were first logged. Numbers D20 and D28 are not used. Unless stated otherwise, figures refer to the analysis sample of 1,727 authority-years across 295 authorities, with City of London excluded (D21). The headline model (Model 2) looks at net current expenditure per resident against IMD score, controlling for region and a COVID-year indicator, with standard errors clustered by authority.

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D1: Authentication method

Date: 24 July 2026 | Stage: Infrastructure setup

Decision: Connect to BigQuery through OAuth rather than a service account keyfile.

Alternative: A service account keyfile. This would make fully automated replication easier, but it means managing a credential that could end up in a public repository.

Why: For a project run by one person, OAuth works well for local development and avoids publishing anything that could give access to the project.

Effect: Anyone reproducing the platform has to set up OAuth themselves. This is a small barrier to reproduction, and the README explains the steps.

Accountability: Information: documented. Explanation: setting up OAuth takes some technical knowledge, which limits who can realistically reproduce the work.

D2: BigQuery dataset region

Date: 24 July 2026 | Stage: Infrastructure setup

Decision: Store the data in the EU multi-region.

Alternatives: The US default region, or europe-west2 (London).

Why: EU multi-region keeps processing within EU jurisdiction, which fits UK public-sector data practice. The first dataset was created in europe-west2, but this clashed with dbt's location setting, so the dataset was recreated in EU multi-region.

Effect: None on the analysis. It is recorded because it shows how an early technical setting can cause problems later.

Accountability: Information: documented. No bearing on Explanation or Consequences.

D3: IMD deprivation measure

Date: 24 July 2026 | Stage: Staging model

Decision: Use the IMD average score as the deprivation measure.

Alternative: The IMD average rank.

Why: The score is continuous, which suits linear regression. A rank treats every step between neighbouring authorities as equal, even when the underlying scores are far apart or almost identical. IMD File 10 provides both, and the score is the usual choice for regression. A higher score means more deprived.

Effect: Results are expressed per point of IMD score, which is what allows the headline finding to be read as "£35.92 more spending per resident for each IMD point".

Accountability: Information: documented. Explanation: a score is harder for a general reader to grasp than a rank, so the dashboard's About page explains it.

D4: Revenue Outturn status filter

Date: 24 July 2026 | Stage: Staging model

Decision: Keep only rows marked "submitted", leaving out the aggregate "total" rows and the "not submitted" rows.

Alternatives: Keeping the total rows, or keeping the not-submitted rows and treating their missing values as zero.

Why: Total rows add up several authorities, so keeping them would count spending twice. A not-submitted row means the figure is missing, not that the council spent nothing.

Effect: The total rows and 23 not-submitted rows were removed. The not-submitted authorities are listed in D11.

Accountability: Information: documented. Explanation: nobody looking at the outputs would know these rows had been removed, so the exclusion is recorded here and on the dashboard's About page.

D5: Service category scoping

Date: 24 July 2026 | Stage: Mart model planning

Decision: Use the RS_ summary variables that MHCLG already publishes, instead of picking from the thousands of individual service-line columns.

Alternative: Decoding the individual columns with the MHCLG data dictionary and choosing service lines by hand.

Why: The individual columns have coded names that can only be read with the data dictionary, and MHCLG provides the RS_ totals specifically for high-level analysis. Using them avoids a whole extra layer of judgement about which lines to include.

Effect: The analysis relies on MHCLG's own groupings. The specific RS_ variable used as the outcome was chosen in D9.

Accountability: Information: documented. Explanation: RS_ names mean little on their own, so the dissertation and dashboard explain them.

D6: COVID year treatment

Date: 27 July 2026 | Stage: Mart model

Decision: Keep 2020-21 and 2021-22 in the headline model and add a COVID-year indicator.

Alternatives: Dropping the COVID years altogether, or keeping them with no indicator.

Why: Dropping them would shrink the sample from 1,727 to 1,147 authority-years. The indicator keeps the full panel while letting average spending in those two years differ from the rest.

Effect: A check that leaves out the COVID years was planned in advance (D15, S1).

Concern at the time: A single indicator assumes COVID affected every authority in the same way. If more deprived authorities received extra emergency funding, this could distort the deprivation result.

What happened: Leaving out the COVID years barely moves the deprivation result (£36.48 against £35.92), so the headline finding does not depend on that period. Separately, the indicator's own negative value turned out to reflect a comparison with later, higher-spending years rather than anything COVID did (D23). Year dummies would be a better way of handling time in future work, and they leave the deprivation result almost unchanged (£35.97).

Accountability: Information: documented, including how the concern played out. Explanation: the dashboard gives no sign of how the COVID years were handled unless a note says so.

D7: Council merger handling

Date: 27 July 2026 (Somerset added September 2026) | Stage: Mart model

Decision: Join Revenue Outturn to IMD 2025 with an inner join. This leaves out the councils that were merged away, while new merged councils are included from the year they were created.

Alternative: Adding up the old councils' spending onto the new boundaries, weighted by population.

Why: IMD 2025 uses 2024 boundaries, so the old councils have no deprivation score. Rebuilding their spending on new boundaries would mean more judgement calls about weighting and would mix figures from different administrative units.

Mergers affected:

- Buckinghamshire, 2020: Aylesbury Vale, Chiltern, South Buckinghamshire, Wycombe.
- North Northamptonshire, 2021: Corby, East Northamptonshire, Kettering, Wellingborough.
- West Northamptonshire, 2021: Daventry, Northampton, South Northamptonshire.
- Cumberland, 2023: Allerdale, Barrow-in-Furness, Carlisle, Copeland.
- Westmorland and Furness, 2023: Eden, South Lakeland.
- North Yorkshire, 2023: Craven, Hambleton, Harrogate, Richmondshire, Ryedale, Scarborough and Selby.

- Somerset, 2023: Mendip, Sedgemoor, South Somerset, and Somerset West and Taunton.

Revised: The first version listed six mergers. Somerset was only spotted later, while building the dashboard, when a gap showed up on the 2022-23 map (D25).

Effect: The number of authorities changes from year to year, because of merger timing and the missing returns in D11. The panel is therefore unbalanced: 270 authorities appear in all six years.

Accountability: Information: documented with codes. Explanation: users would not know the old councils are missing unless told, so the dashboard's coverage notes name them.

D8: Authority count

Date: 24 July 2026 | Stage: Data inspection

Decision: Work with the 296 local authority districts in IMD 2025 File 10, correcting the figure of about 317 given in the proposal.

Alternative: Looking into whether 317 could be right under a different geography, for example by including county councils.

Why: File 10 states that England had 296 local authority districts when it was published, and these lower-tier districts are the right unit for this analysis.

Effect: The analysis starts from 296 districts and, after City of London is removed (D21), covers 295. Other bodies in the Revenue Outturn (county councils, police, fire and combined authorities) drop out through the inner join.

Accountability: Information: documented. The error was caught while inspecting the data, before any analysis.

D9: Primary spending variable

Date: 27 July 2026 | Stage: Mart model

Decision: Use net current expenditure (RS_netcurr_tot_net_exp, line 749) as the main outcome. Total service expenditure (RS_tot_sx_net_exp) is used as a check, and adult and children's social care (RS_asc_net_exp, RS_csc_net_exp) as service-level checks.

Alternative: Using total service expenditure as the main outcome.

Why: MHCLG defines net current expenditure as line 749, which is total service expenditure plus financing items such as levies and trading-account adjustments. The funding literature cited (Innes and Tetlow, 2015; Amin-Smith and Phillips, 2019; Fahy et al., 2023) uses the term in the same way, so calling total service expenditure "net current expenditure" would have been misleading.

Checks carried out:

- The line 749 definition was confirmed in MHCLG's statistical release.
- An external data specialist recommended the same choice.
- The size of the difference between the two measures was only weakly related to IMD score.
- The difference was much larger for Shire Districts than for other authority types. Shire Districts do not run social care (a county responsibility), so their service spending is small compared with the financing items included in net current expenditure. The gap is therefore about

authority type rather than deprivation, which is the same pattern that later explained the residuals in D26.

Effect: Across the analysis sample, total service expenditure averages £825 per resident, against £1,066 for net current expenditure. Results of the checks are in D15.

Accountability: Information: documented with the checks above. Explanation: the difference between the two lines is technical, so it is explained on the dashboard's About page and in Section 9.1.

D10: Column name standardisation

Date: 24 July 2026 | Stage: Ingestion

Decision: Convert all column names to snake_case with a small regex function before loading to BigQuery.

Alternatives: Mapping names through a BigQuery schema, or shortening them.

Why: BigQuery does not accept spaces, brackets or special characters in column names, and the IMD file has all three (for example "Local Authority District code (2024)"). Snake_case keeps the names readable, which shortened names would not.

Effect: All dbt models and Python code use snake_case names. The function is in the ingestion notebook.

Accountability: Information: documented. This was a technical necessity rather than an analytical choice.

D11: Not-submitted returns

Date: 24 July 2026 | Stage: Data inspection and staging model

Decision: Leave out returns that were never submitted, and list every authority affected.

Alternatives: Filling the gaps from earlier years, treating them as zero, or using MHCLG's grossed-up estimates.

Why: Filled-in values could not be checked against anything, and zeros would make it look as if these councils spent nothing. MHCLG's grossed-up estimates cover England as a whole, not individual authorities.

Effect: 23 rows were removed, all from the last two years:

- 2023-24 (14 rows): Birmingham, Westmorland and Furness, Cambridge, Brentwood, Castle Point, Colchester, Pendle, North West Leicestershire, Cannock Chase, Stafford, Babergh, Mid Suffolk, Woking, and North of Tyne Combined Authority. The last is not a district and would have been outside the panel anyway.
- 2024-25 (9 rows): Birmingham, Slough, Cumberland, Amber Valley, North West Leicestershire, Ashfield, Guildford, North Warwickshire and Warwick.

Birmingham, England's largest authority and a highly deprived one, is missing from both years, which matters when comparing 2023-24 and 2024-25 with earlier years. A check on the 270 authorities present in all six years is reported in D23.

Accountability: Information: documented with the full list. Explanation: dashboard users would not know these authorities were missing, so the About page names Birmingham and mentions the rest. Consequences: nothing requires councils that miss the deadline to provide their data.

D12: Population denominator

Date: 27 July 2026 | Stage: Mart model

Decision: Divide spending by the ONS mid-year population for the matching year: mid-2019 for 2019-20, through to mid-2024 for 2024-25.

Alternatives: One population figure for all years, an average across the period, or the population figures in the Revenue Outturn file.

Why: Matching the year is the most accurate option when populations change. A single base year would make spending per resident look too high in growing areas in later years, and too low in shrinking ones.

Source: ONS mid-year population estimates, MYE4 sheet (mye24tablesuk.xlsx), downloaded August 2026.

Effect: Per-resident figures reflect each year's population. Mid-2025 estimates were not yet out, so mid-2024 figures are used for 2024-25, which slightly understates population in fast-growing areas.

Accountability: Information: documented. Explanation: readers might assume one fixed population figure was used, so the method is stated.

D13: IMD duplicate rows

Date: 1 August 2026 | Stage: Data quality testing

Decision: Reload raw_imd_2025 with an overwrite setting (WRITE_TRUNCATE) to clear duplicated rows, and use the same setting for every raw load from then on.

What went wrong: The IMD table had been loaded more than once in earlier sessions without overwriting, so 10 authorities appeared twice.

How it was found: The dbt uniqueness test on stg_imd_2025.lad_code failed.

How it was fixed: The table was reloaded with 296 unique rows and the mart rebuilt. Its row count fell from an inflated figure to 1,733.

Effect: Without the fix, the duplicated authorities would have counted twice in the regression, and nothing in the output would have looked wrong.

Accountability: Information: an automated test caught a real error before the analysis was run. Explanation: a technical reader can follow the fix through this record and the dbt tests, but a dashboard user would never see it.

D14: GitHub repository configuration

Date: 1 August 2026 | Stage: Publication

Decision: Publish the code and documentation in a public repository (<https://github.com/kleaa-hash/deprivation-spending-analysis>). Credentials (profiles.yml) and generated folders (logs/, target/)

are kept out through .gitignore. This catalogue is published in the repository as docs/decisions_catalogue.pdf.

Alternatives: A private repository, or separate repositories for code and documentation.

Why: Objective 5 requires public access, and it supports the Information dimension. Credentials are left out for security, and generated files are not source code.

Effect: Anyone with their own Google Cloud account and OAuth setup can rerun the pipeline and analysis by following the README. The Looker Studio dashboard is linked from the repository, but its report settings cannot be stored there.

Accountability: Information: public access. Explanation: reproducing the work takes technical knowledge, which limits who can use this route.

D15: Regression specification and sensitivity analyses

Date: 1 August 2026; results 23 August 2026; revised September 2026 | Stage: Analysis

Decision: Pooled OLS across all six years, with net current expenditure per resident as the outcome and IMD score as the main explanatory variable. The original plan controlled for the COVID-year indicator and authority type; region was added in D18. Standard errors are clustered by authority. The headline model (Model 2) leaves out authority type (see D22).

Alternatives:

- A separate regression for each year. Rejected because each year has fewer than 300 authorities.
- A multilevel or fixed-effects model. Left for future work, because fixed effects would wipe out the IMD score, which does not change over time (D16).

Criteria set before running: A positive relationship would be accepted if the IMD coefficient was positive and significant at $p < 0.05$. It would be treated as weak or absent if it was not significant, or if R-squared was below 0.10. Either outcome would be reported.

Checks planned in advance: (S1) leaving out the COVID years; (S2) total service expenditure as the outcome; (S3) adult social care per resident; (S4) children's social care per resident.

Results:

- Model 2: £35.92 per IMD point ($p = 5.89 \times 10^{-11}$, R-squared = 0.562, $n = 1,727$). Meets the criterion.
- S1, without the COVID years: £36.48 ($p < 0.001$, R-squared = 0.556, $n = 1,147$). The result does not depend on the COVID period.
- S2, total service expenditure: £28.17 ($p < 0.001$, R-squared = 0.500). This is about 22% lower, which roughly matches the smaller size of the measure, so the relationship holds in proportion under either definition.
- S3, adult social care: £0.57 ($p = 0.33$, R-squared = 0.182, $n = 760$, 131 authorities). Does not meet the criterion.
- S4, children's social care: £5.25 ($p < 0.001$, R-squared = 0.347, $n = 760$, 131 authorities).

Revised: S3 and S4 were first run on all authorities, and both results were significant. But Shire Districts do not provide social care, and their recorded spending on it is close to zero. The first run

was therefore mostly comparing councils that provide social care with councils that do not. S3 and S4 were rerun on social-care authorities only (London Boroughs, Metropolitan Districts and Unitary Authorities). This restriction was added after the first run rather than planned in advance. The adult social care result did not hold up. Earlier statements that "all four sensitivity analyses confirmed the result" are withdrawn.

Accountability: Information: the plan, criteria and all results are recorded, including the revision.
Explanation: the results differ across the checks because each one answers a slightly different question, and that needs explaining.

D16: Pooled rather than fixed-effects regression

Date: 1 August 2026 | Stage: Analysis planning

Decision: Pooled OLS with COVID-year and regional controls, rather than a panel model with authority fixed effects.

Alternatives: Authority fixed effects, or first differences.

Why: Each authority has the same IMD 2025 score in every year (D17). Fixed effects would absorb it completely, so the main result could not be estimated at all. First differences would throw away the differences in levels between authorities, which is exactly what the research question is about.

Effect: The models describe patterns; they do not show cause and effect, because hidden differences between authorities are not controlled for. Since the same authority appears in several years, standard errors are clustered by authority. A study using several editions of the IMD could support a proper panel design.

Accountability: Information: the limitation is documented. It also limits what the platform can say about change over time.

D17: Time-invariant IMD

Date: 1 August 2026 | Stage: Analysis planning

Decision: Apply IMD 2025 to all six years, from 2019-20 to 2024-25.

Alternatives: IMD 2019 for the earlier years and IMD 2025 for the later ones, or IMD 2019 throughout.

Why: IMD 2025 is the most up-to-date measure. Switching editions partway would create a break in the data, and IMD 2019 would be less accurate for the later years.

Effect: The analysis assumes deprivation stayed roughly the same over the period. The fit is likely to be weakest for the earliest years, which come before the index was compiled. Because each authority's score is unique and never changes, it also works as an authority identifier in the Random Forest (D24). The proposal planned a check using IMD 2019, but this was not carried out. It would have had limits anyway, since the 2025 edition added or substantially revised 34 indicators. The gap is recorded as a limitation.

Accountability: Information: documented. It limits what the platform can say about deprivation changing over time.

D18: Regional control variable

Date: 1 August 2026 | Stage: Mart model and analysis

Decision: Add ONS region (nine regions) as a control.

Source: ONS local authority district to region lookup (December 2024), 296 authorities.

Alternative: No regional control.

Why: Spending differs between regions for reasons that have nothing to do with deprivation, such as London's higher costs and historical funding levels. Region and deprivation are also linked, so without a regional control the model could credit regional differences to deprivation.

Effect: Region is included in every model, including the headline Model 2. Authority type was dealt with separately (D22).

Accountability: Information: documented. The choice changes what the deprivation result means, which Section 9.1 explains.

D19: Negative net current expenditure (Spelthorne)

Date: 23 August 2026 | Stage: Analysis

Decision: Keep Spelthorne's negative net current expenditure in the analysis.

Alternatives: Removing Spelthorne, or replacing the negative values with zero.

Why: A negative figure means the council's income was higher than its spending that year. In Spelthorne's case this comes from commercial income, not from a data error. Removing or changing the values would misrepresent what was actually recorded.

Effect:

- Spelthorne is the only authority with negative net current expenditure, in four of its six years (2019-20 to 2022-23). Its figures are positive in 2023-24 and 2024-25 (D31).
- Leaving Spelthorne out makes almost no difference to the deprivation result (£35.88, $p = 6.30 \times 10^{-11}$, R-squared = 0.562).
- None of Spelthorne's rows is among the high-influence rows flagged by Cook's distance (D21).
- The four negative values cannot be logged, so they are left out of the log-spending trend models only.

Accountability: Information: documented. Explanation: negative spending would puzzle dashboard users, so the dashboard includes a note.

D20: Not used

The Spelthorne check first logged under this number is now part of D19.

D21: City of London exclusion

Date: 24 August 2026 | Stage: Regression diagnostics

Decision: Leave City of London out of all models.

Alternative: Keeping it in.

Why: City of London has very few residents but very high spending linked to its commercial role, so its per-resident figures are extreme. Cook's distance for its six rows is 53.4 to 98.7 times the usual $4/n$ threshold ($n = 1,733$).

Effect: With City of London included, the deprivation result falls to £15.59 ($R^2 = 0.229$) and is no longer significant under the clustered standard errors used throughout ($p = 0.447$). The headline finding therefore depends on this exclusion, and Section 9.1 says so plainly.

Also found: Cook's distance flags three of the six Isles of Scilly rows, slightly above the threshold. No other authority has rows above it. See D29.

Accountability: Information: documented with the diagnostics. Explanation: the fact that the result rests on one exclusion needs to be said openly, not left for readers to discover.

D22: Authority type and the headline model

Date: 23 August 2026; revised September 2026 | Stage: Regression diagnostics

What was found: Model 1, which includes both authority type (`la_class`) and region, is rank-deficient (condition number 1.10×10^{16}). The variance inflation factors for `region_name_London` and the three non-baseline `la_class` dummies are effectively infinite, while all other variables sit between 1.0 and 2.3. The cause is built into the data: every London authority is both in the London region and a London Borough, so `region_name_London` can be written exactly in terms of the `la_class` dummies. This only affects how the London effect is split between the two variables. The IMD result, fitted values and residuals are not affected, which a full-rank refit confirmed (largest difference in residuals 3×10^{-12}).

Decision: Report Model 2, without authority type, as the headline result, with Model 1 shown alongside it.

Original reasoning, since replaced: The entry first said the problem affected London only, that `la_class` had to be dropped to fix it, and that Model 1 would be kept as a "Model 1a" check. An earlier draft of the findings also described Model 2 as the more reliable estimate.

Revised:

- The problem involves all three `la_class` dummies together with `region_name_London`, not London alone.
- Authority type did not need to be dropped, because Model 1 is still valid.
- Model 2 is not more reliable. It is the headline because it describes the deprivation gradient across authorities as they actually are (£35.92 per IMD point), while Model 1 describes the gradient within each authority type (£13.98). Adding authority type cuts the gradient by about 61% and raises R^2 from 0.562 to 0.918.

Effect: Model 2's result is read as a picture of the system as it stands: whether spending follows deprivation in practice, not after allowing for differences between types of council.

Accountability: Information: the finding, the original reasoning and the correction are all kept. Explanation: the difference between the two models is too subtle for chart captions, so Section 9.1 and this record explain it.

D23: Interpretation of the COVID coefficient

Date: 23 August 2026; revised September 2026 | Stage: Findings

What was found: The COVID-year result in Model 2 is negative: £56.07 less per resident in 2020-21 and 2021-22 ($p < 0.001$), even though emergency funding was paid in those years.

Original explanation, since replaced: It was thought that COVID grants had been recorded as income in high-spending authorities, which would reduce their net current expenditure. The evidence given was that average spending was lower in the COVID years (£1,024 against £1,087) while the median was higher (£432 against £407).

How it was tested: Model 2 has no other time controls, so the indicator is really comparing the COVID years with the other four years combined, three of which (2022-23 to 2024-25) had the highest spending in the panel.

- With a dummy for each year, spending in both COVID years is higher than in 2019-20, the only year before COVID, by £71 and £63 per resident ($p < 0.001$).
- With a linear year trend, the COVID result becomes small and positive (£8.21, $p = 0.04$).
- On the 270 authorities present in all six years, the result is still negative, so it is not caused by changes in which councils reported.
- Between 2019-20 and 2024-25, average spending rose by £231 per resident but the median by only £40, which explains the gap between mean and median in the pooled comparison.

Revised conclusion: The negative result comes from what the indicator is being compared with, not from COVID itself. With only one year before COVID and spending rising throughout, the data cannot separate out a COVID effect in either direction. The income explanation was never tested directly, so it is neither confirmed nor ruled out. The COVID result is not interpreted further.

Effect: The deprivation result barely changes with year dummies (£35.97 against £35.92). Year dummies would be the better way of handling time in future work.

Accountability: Information: the original explanation, the test and the revision are all recorded.
Explanation: an explanation that seemed obvious turned out to depend on how time was modelled.

D24: Random Forest specification and validation

Date: 23 August 2026; revised September 2026 | Stage: Machine learning

Decision: Train a Random Forest on the same inputs as Model 2 (IMD score, COVID-year indicator and region dummies), predicting net current expenditure per resident. It uses 500 trees and otherwise default settings, with no tuning.

Alternative: Adding authority type as an input, since tree models are not affected by collinearity the way OLS is.

Why: Using the same inputs as Model 2 makes the comparison with the linear model fair.

How validation developed:

- A random 80/20 split gave a test R-squared of 0.935. This is inflated by leakage. Each authority's IMD score is unique and fixed across years (D17), so it identifies the authority, and a random split puts other years of the same authority into the training data.

- Five-fold cross-validation without shuffling gave a mean R-squared of -43.55. Revised: this was first blamed on the same leakage, but a direct test showed a different cause. The data is sorted by authority code, and the code prefixes follow authority type, so the unshuffled folds ended up as near-uniform blocks. One fold was made up entirely of Shire Districts (all 59 of its authorities) and scored an R-squared of -183.05.
- Five-fold cross-validation with shuffling gave a mean R-squared of 0.968. This fixes the sorting problem but not the leakage.
- Group K-Fold by authority gave a mean R-squared of 0.276 (fold range 0.10 to 0.34). This is the only one of the four that tests the model on authorities it has never seen. The folds had a good mix of authority types (no single type made up more than 58% of any fold, against 56% overall). Model 2 on the same folds averages an R-squared of 0.546 (range 0.50 to 0.65), better in every fold and close to its in-sample 0.562.

SHAP: SHAP values were calculated with TreeExplainer, which is exact for the model it explains, across all 1,727 rows. They explain the model trained on the random split, so they describe the model affected by leakage. IMD score comes first (mean |SHAP| 452.6), followed by region_name_London (288.8). IMD score's lead is at least partly because it acts as an authority identifier. SHAP values were not recalculated for the Group K-Fold models.

Revised: Earlier notes that treated the Random Forest's feature importances as a useful guide are withdrawn, because they also come from the model affected by leakage.

Effect: With these inputs and no tuning, the linear model does a better job on authorities it has not seen. The Random Forest's apparent strength came almost entirely from how it was tested.

Accountability: Information: the full sequence is recorded, including the mistaken explanation. Explanation: a precise explanation (SHAP) of a model that does not generalise gives no warning that anything is wrong (Section 9.4).

D25: Representative year for the maps

Date: 24 August 2026; revised September 2026 | Stage: Spatial analysis

Decision: Use 2022-23 for the single-year maps (IMD score, spending per resident and Model 2 residuals). A second residual map shows each authority's average residual across all the years it appears.

Alternative: The most recent year, 2024-25.

Why: No year has full coverage. 2022-23 avoids both the COVID years and the missing returns of 2023-24 and 2024-25 (D11). It covers more authorities than 2023-24, and its gaps are few, close together and easy to explain.

Coverage of the single-year maps: 2022-23 is missing Cumberland, Westmorland and Furness, North Yorkshire and Somerset, all created in 2023. For that year the Revenue Outturn still reports these areas under the old councils' codes, while IMD 2025 only has the new codes, so neither side matches (D7). Revised: Somerset was left off this list at first and was found while building the dashboard.

Coverage of the average map: All 295 authorities appear, but the four 2023 authorities are based on only one or two years: Cumberland and Westmorland and Furness on one year each, North Yorkshire on two, and Somerset on one or two. Every other authority is based on up to six.

Effect: The map captions name the four missing authorities and flag the short averages. The residual figures in Section 9.1 are averages across all years, COVID years included; the single-year map is only used to show the geographic picture.

Accountability: Information: documented. Explanation: a blank area or a short average would easily be misread, so the dashboard states it on the map itself.

D26: Authorities at the residual extremes

Date: 24 August 2026; revised September 2026 | Stage: Findings

Decision: Before using the authorities furthest from Model 2's predictions as examples, check whether the pattern still holds once authority type is added (Model 1's specification). Figures are average residuals across each authority's years in the panel, in £ per resident.

Lowest residuals (spending less than predicted): All ten are Shire Districts. Eight are in the North West: Burnley, Pendle, Hyndburn, Preston, Rossendale, Lancaster, Wyre and West Lancashire. Each of these is below prediction in every year it appears. Once authority type is added, the ten residuals shrink by 84% to 93%, from between -£873 and -£1,489 to between -£67 and -£243.

How much region contributes: Measured against a model with neither region nor authority type, region explains roughly 10-13% of the shrink for three of the North West authorities. For the other five it works in the opposite direction.

Revised: Earlier versions named only seven North West authorities (Wyre was missed), and an earlier run of the region comparison was affected by an index-alignment error. Both were corrected in a clean rebuild.

Highest residuals (spending more than predicted): All ten are Unitary Authorities. Isles of Scilly has the largest residual (£2,875) and shrinks by only 30%, which fits with its very small population (D29). The other nine shrink by 69% to 99%. In three cases the apparent overspend becomes an underspend: Wokingham (£1,271 to -£76), Rutland (£1,269 to -£133) and Central Bedfordshire (£1,203 to -£121). Bedford falls close to zero (£1,184 to £6). Wokingham's population (171,789 to 187,200) is normal for a Unitary Authority, so a small population does not explain its result.

A possible explanation, not tested: Unitary Authorities may look as if they overspend because many of the councils they are compared with in their region are Shire Districts, which provide fewer services.

Note on method: Model 1 is rank-deficient (D22), but its residuals are still uniquely determined, so they can be compared with Model 2's.

Effect: Neither group is presented as a case of geographic or deprivation-related under- or overspending. Instead, Section 9.1 uses both as evidence that most of what Model 2 cannot explain comes from authority type rather than geography, in line with R-squared rising from 0.562 to 0.918 once authority type is added. Residuals show statistical patterns; they are not judgements on individual councils.

Accountability: Information: figures and revisions documented. Explanation: the map on its own suggests a geographic story; only the authority-type check shows what is really behind it.

D27: Residual map colour scale

Date: 24 August 2026 | Stage: Spatial analysis

Decision: Use a two-sided colour scale centred on zero for the residual maps, with red above prediction and blue below.

Alternative: A one-directional scale running from the lowest residual to the highest.

Why: Zero means spending exactly as predicted. The largest positive residuals are much further from zero than the largest negative ones, so a one-directional scale would put the neutral colour somewhere other than zero and give a false impression of who is above or below prediction.

Effect: Pale colours mean close to prediction. Each side of the scale stretches to its own most extreme value, so the same shade of red and blue does not mean the same amount; the colour bar shows the actual figures.

Accountability: Information: documented. Explanation: a design choice can change how results are read, so it is recorded in the same way as an analytical one.

D28: Not used

D29: Isles of Scilly sensitivity test

Date: 24 August 2026 / Stage: Analysis

Decision: Keep the Isles of Scilly in the headline model, and check its influence by rerunning the model without it.

Why: Its very small population pushes per-resident figures up, and Cook's distance flags three of its six rows (D21).

Effect: Without the Isles of Scilly, the result moves from £35.92 to £36.97 (about 3%) and R-squared from 0.562 to 0.589, with the direction and significance unchanged ($p < 0.001$). The headline model does not depend on including it.

Accountability: Information: documented.

D30: Static choropleth maps

Date: 2 September 2026 / Stage: Dashboard construction

Decision: Make the maps as static images in Python and place them in the dashboard, instead of using Looker Studio's own map charts.

Why: Looker Studio's geo chart, filled map and bubble map would not accept English local authority district boundaries. All three were tried and each gave an "invalid dimension" error.

Effect: The maps are not interactive, which in turn limits D31.

Accountability: Information: documented. Explanation: viewers would not know the maps are static out of necessity unless told.

D31: Year filter on the per-year scatter chart

Date: 4 September 2026 / Stage: Dashboard construction

Decision: Add a year selector to the deprivation-versus-spending scatter chart only, not to the whole dashboard.

Why: The maps are static images (D30) and cannot respond to a filter. A dashboard-wide filter would change some pages and silently leave others as they were.

Checked before building: The chart caption was checked against every year, not just 2022-23:

- The unadjusted slope is positive in every year.
- Shire Districts form the lower band in every year.
- The Isles of Scilly is among the three highest spenders in every year.

One statement did not hold. Spelthorne's spending is negative from 2019-20 to 2022-23 but positive in 2023-24 and 2024-25, so a fixed "below zero" label would have been wrong for two of the six years.

Effect: The caption now says Spelthorne is negative in four of the six years, so where it appears depends on the year selected. The caption is correct for every year a viewer can pick.

Accountability: Information: documented, including the check. Explanation: even a small addition to the dashboard had to be checked to keep it accurate.